



HP Enterprise - Quantum Resistant (QR) Enablement Process

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Overview

Quantum computing will break today's NIST (National Institute of Standards and Technology) standard asymmetric encryption-based cryptography (e.g., RSA, Elliptic Curve) in minutes, undermining digital signatures and firmware integrity. This creates a "harvest-now, decrypt-later" risk where attackers store encrypted data today and decrypt it once quantum machines mature.

The U.S. Federal Government has committed that starting in **2027**, all devices for National Security Systems must use quantum-resistant digital code signing for firmware and software. This aligns with NSA's CNSA 2.0 guidance and NIST's finalized PQC standards (ML-DSA : NIST FIPS PUB 204 (Section 5.1) ML-KEM : NIST FIPS PUB 203 (Section 7.1)).

HP introduced the **world's first quantum-resistant printers** in our HP LaserJet Enterprise 8000 Series and Managed E700/E800 Series in addition to our HP Color LaserJet Enterprise 5000/6000 and X500/X600 Series, featuring both hardware and firmware layers that enable quantum resistance at the device level.

Quantum-resistant printers complement **HP's Zero Trust Architecture**, enabling secure device identity and firmware integrity checks across fleets. This ensures resilience against lateral movement attacks even in hybrid environments.

System requirements

- **Firmware:** HP FutureSmart 5.9.2.3 or newer.
- **Hardware:** Quantum Resistant-capable A3 formatter (PCA code ending in 13–18) and A4 formatter with an updated revision number listed below.
NOTE: If a supported formatter is not detected, the quantum resistant option will not be visible via the configuration page or security menu within the EWS.
- **Browser Access:** Ensure you can access the printer's Embedded Web Server (EWS) via a supported web browser.
- **Admin Rights:** You must have administrator credentials for the device in order to enable the quantum-resistant feature.

Limitations

After enabling quantum resistance, downgrading to any **non-Quantum Resistant (QR) firmware version is not allowed**. Only downgrades between different quantum-resistant firmware versions are allowed. This is a deliberate security measure to prevent downgrade attacks and maintain cryptographic integrity.

NOTE: This feature requires manual enablement for the products listed below. Future products will have this feature enabled by default, active in all Enterprise and select business-class Pro devices from HP factories.

Enable the Quantum Resistant (QR) feature

To enable quantum resistant (QR) via the supported device, follow these steps.

1. Open a web browser and enter the printer's IP address.
2. Log in using administrator credentials.
3. Select the **Security** menu.
4. Select **General Security** from the left pane.
 - a. Find the "Quantum Resistant Secure Boot Activation" section.

Quantum Resistant Secure Boot

The Quantum Resistant Secure Boot is currently disabled.

Enable

5. Click the **Enable** button.

NOTE: A warning will display stating this action cannot be reversed or disabled.



Activate Quantum Resistant Secure Boot

This operation will activate a feature designed to protect the firmware against potential quantum computer attacks. Click "Activate" to continue. Activating this feature will permanently enable Quantum Resistant Secure Boot.

Once activated, it cannot be reversed or disabled. Only firmware bundles that support this feature can be installed.

6. Click "**Activate**" to confirm.



The product is shutting down to complete the operation.

Please wait a few minutes, then refresh this page to see if the product has restarted. This web server will not be available until the product restarts.

7. Verify the status on the Configuration page in the Embedded Web Server (EWS), Security Menu or via the printer's control panel.

Security

Hard Disk Encryption Status:

Internal Disk:	Encrypted
Job Data Encryption Status:	Encrypted
Job Data Persistence:	Persistent (not volatile)
File Erase Mode:	Non-Secure Fast Erase (No overwrite)
Control Panel Password:	Disabled
Support Key:	V6GD-RSFG-TW5L
Host USB plug and play:	Enabled
Device USB:	Enabled
HP SureStart:	Present (Quantum Resistant: Configured)
Whitelisting:	Development
Runtime Intrusion Detection	Present



Quantum Resistant Secure Boot

Quantum Resistant Secure Boot has been enabled on this product and cannot be disabled. Only firmware bundles that support this feature can be installed. For information about firmware versions, visit the Software and Drivers website at <https://support.hp.com/>

Table 1. Formatter Part Numbers (A3 PCA Number)

Product Model	Unsupported Formatter		Supported QR Formatter	
	Unsupported Formatter	Supported QR Formatter	Formatter PCA Part number	Service Part Number
E/D826xx, 8501 & 8601	3SJ33-60001	6CF14-67002	3SJ33-60013	6CF14-67012
E/D877xx & 8801	3SJ33-60002	6CF14-67001	3SJ33-60014	6CF14-67011
E/D731xx	3SJ00-60001	6CF14-67004	3SJ00-60015	6CF14-67014
E/D786xx	3SJ00-60002	6CF14-67003	3SJ00-60016	6CF14-67013
E730xx	3SJ00-60003	6CF14-67006	3SJ00-60017	6CF14-67016
E785xx	3SJ00-60004	6CF14-67005	3SJ00-60018	6CF14-67015

NOTE: The PCA part numbers listed above are formatter board identifiers and are not orderable. For service replacement, always use the corresponding Service Part Number.

Table 2. Formatter Part Numbers (A4 Revision Number)

Note: The highlighted products in **red** are unavailable for quantum resistant upgrade until a later date.

Product Model	Unsupported Formatter	Supported QR Formatter	Supported QR Service Part Number
5700, X/D 55745xx	D6423.02 & E6430.00	F6510.00	6QN27-67002
5800, X/D 579xx, X580xx	C6315.01	D6507.00	6QN29-67002
6700 & X654xx	C6334.00	D6507.00	6QN32-67002
6800 & X/D 677xx	C6315.01	D6507.00	6QN35-67002

Supported Printers

A3 Products Model Numbers and Product Numbers

- **HP LaserJet Managed MFP E730xx**
 - 5QJ87A, 3SJ03A, 3SJ04A
- **HP Color LaserJet Managed MFP E785xx**
 - 5QJ83A, 5QJ81A, 5QK15A

- **HP Color LaserJet Managed MFP E786xx**
 - 5QJ90A, 5QJ94A, 5QK18A, 3SJ11A, 3SJ12A, 3SJ13A, 3SJ32A, 3SJ33A, 3SJ34A, A91S5A, A91S4A
- **HP Color LaserJet Managed MFP E877xx**
 - 5QK03A, 5QK08A, 5QK20A, 3SJ19A, 3SJ20A, 3SJ21A, 3SJ22A, 3SJ35A, 3SJ36A, 3SJ37A, 3SJ38A, A91S6A, A91S7A
- **HP Color LaserJet Ent Flow MFP and MFP 8801x**
 - 9S183A, 9S184A, AQ1E2A, BD5G8A
- **HP LaserJet Managed MFP E826xx**
 - AG8Z1A, A91S3A, A91S2A, 5QK09A, 5QK13A, 5QK21A, 3SJ07A, 3SJ08A, 3SJ09A, 3SJ28A, 3SJ29A, 3SJ30A
- **HP Color LaserJet Ent Flow MFP and MFP 8501x & 8601x**
 - 9S187A, AJ7J3A, AQ1E4A, BD5H0A, 9S185A, 9S186A, AQ1E3A, BD5G9A
- **HP LaserJet Managed MFP E731xx**
 - 5QJ98A, A91S1A, A91S0A, 5QK02A, 5QK19A, 3SJ00A, 3SJ01A, 3SJ02A, 6BS57A, 6BS58A, 6BS59A

A4 Products Model Numbers and Product Numbers

- **HP Color LaserJet Enterprise 5700 & HP Color LaserJet Enterprise X55745xx**
 - 6QN28A, 6QP97A, A8ZQ5A, 9Z0H9A
- **HP Color LaserJet Enterprise MFP 6800 & HP Color LaserJet Enterprise MFP X677xx**
 - 6QN35A, 6QN36A, 6QN37A, 6QN38A, 76H10A, 76H11A, 6QQ01A, 6QQ02A, 6QQ03A, 76H08A, 76H09A, 49K86A, 49K88A, 49K87A, 49K90A, 49K91A, 49K92A, A8ZQ7A
- **HP Color LaserJet Enterprise MFP 5800 & HP Color LaserJet Enterprise MFP X579/580xx**
 - 6QN29A, 6QN30A, 58R10, 6QP98A, 6QP99A, 76H07A, A8ZQ6A, 7E357A, 7H5W6A, 7H5W7A
- **HP Color LaserJet Enterprise 6700 & HP Color LaserJet Enterprise X654xx**
 - 6QN33A, 58M42A, 6QQ00A, 49L02A, 49L04A

More Information

The activation process modifies the OTP (One-Time Programmable) memory and hardware fuses, making rollback technically infeasible.

Labels outside of product packaging will be marked with a “PQC” (Post Quantum Cryptography) label as shown below:



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